INTELLIGENT RGB IMAGE-BASED SYSTEM FOR EARLY DETECTION AND DIFFERENTIATION OF TEA LEAF DISEASES AND NUTRIENT STRESSES

Project ID: RP[25-26J-178]

Project Proposal Report

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Sri Lanka Institute of Information Technology
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Declaration page of the candidates & supervisor

We declare that this is our own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Co-Supervisor: Date: 29/08/2025
(Ms. Chathurya Kumarapperuma)

Abstract

Tea is a most economically relevant and grown crop in the world. Its productivity and quality are often affected by various biotic stresses (red rust, anthracnose, etc.) and abiotic stresses (potassium deficiency, leaf sunburn, etc.). Identifying them on time and correctly is critical for effective crop management. However, farmers' traditional visual diagnosis is subjective, time-consuming, and prone to misidentification because several stresses can visually present very similar symptoms.

This paper proposes cost-effective, RGB image-based deep learning for detection, discrimination, and measurement of different stresses in tea plants. The method uses low-cost RGB images acquired using smartphones or digital cameras under real-field conditions with preprocessing, data enhancement, and convolutional neural networks (CNNs) using transfer learning. The system classifies tea leaves into categories like healthy, red rust, and potassium deficiency with the additional optional lesion detection and severity estimation module through object detection or segmentation models. The proposed approach emphasizes pragmatic, student-friendly, and scalable solutions that can be implemented in real tea plantations without utilizing expensive hyperspectral equipment. The model's performance is evaluated in terms of accuracy, precision, recall, F1-score, and confusion matrix metrics, and predictions are compared with expert visual diagnosis to verify.

The expected outcomes are high-accuracy stress classification, quantification of lesions, and actionable insights to farmers, allowing timely intervention and improved crop management. The research bridges the gap between laboratory-based disease detection methods and field-level real-world applications, providing a low-cost, non-invasive, yet effective early stress detection system for tea plants with potential integration with IoT devices or UAV-based monitoring systems in the future.

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1.Introduction

1.1 Background on Tea Leaf Health and Stress Detection.

This study is directed towards developing an inexpensive, feasible system for the identification and differentiation of stresses in tea plants with emphasis on biotic stresses like red rust and abiotic stresses like potassium deficiency. The majority of these stresses have the tendency to show similar-looking visual symptoms, such as yellowing, discoloration, or lesions of leaves, that make normal visual diagnosis by farmers challenging and prone to error [1]. To overcome this, the suggested system utilizes low-cost RGB imaging, which can be captured using normal smartphones or digital cameras directly in tea gardens, under actual field conditions with varying lighting, background complexity, and environmental conditions, thereby ensuring a representative dataset [2].

1.2 Challenges of Visual Diagnosis in Tea Agriculture

The collected images are preprocessed to enhance model learning and robustness. Preprocessing steps include resizing, normalization, and data augmentation methods such as rotation, flip, brightness and contrast adjustments, and minor color jittering to simulate real-world variations. These steps enable the deep learning model to generalize to different scenarios [3]. Lesions or stress-affected areas can be labeled using open-source software such as Label Image, with the possibility of classification and optional lesion localization for severity analysis [4].

A deep learning approach is employed by applying transfer learning to pre-trained CNNs such as ResNet50, MobileNetV2, or EfficientNet. Such convolutional neural network (CNN)-based methods have shown great success in plant disease detection across different crops, including tomatoes [5], and various field crops [6]. In the case of tea, recent studies have demonstrated the potential of deep learning approaches for disease classification [7]. For lesion-level detection, object detection algorithms such as Faster R-CNN or Mask R-CNN can be employed to precisely identify the location of stressed regions, while image segmentation techniques allow quantification of the percentage of leaf area affected—providing a valuable measure of stress severity for practical field recommendations [8].

The system workflow consists of:

1. Data Collection: Capturing diverse RGB images from the field.
2. Preprocessing: Enhancing images and augmenting the dataset.
3. Model Training: Fine-tuning pre-trained CNNs on labeled data.

4. Evaluation: Measuring performance using metrics such as accuracy, precision, recall, F1-score, and confusion matrices.
5. Optional Deployment: Applying the trained model for lesion segmentation, severity estimation, or large-scale monitoring.

This solution is designed to be scalable and flexible, with potential future extensions such as IoT-based real-time monitoring or UAV-mounted imaging for large tea gardens. By leveraging accessible equipment and open-source software platforms such as TensorFlow, PyTorch, OpenCV, and Google Colab for GPU-based training, the system is maintained at low cost to students and researchers while ensuring high accuracy and reliability.

1.3 Significance and Future Prospects

The approach presents a non-invasive, high-throughput, and feasible alternative for the early detection and control of tea plant stresses, facilitating knowledge-based decision-making and improved productivity under real-world agricultural conditions. Moreover, the development of such an intelligent system aligns with the growing need for smart agriculture and precision farming technologies aimed at increasing crop yields, reducing input costs, and ensuring environmental sustainability[9]. Computer vision and artificial intelligence have already been shown to enhance cultivation practices by providing growers with timely, data-driven recommendations that reduce reliance on manual inspection and professional expertise [10].

Not only does this system facilitate early stress identification, but it also enables optimization of fertilizer application, pest management strategies, and irrigation scheduling based on the nature of the detected stress. Furthermore, this research contributes to the broader scientific community by providing an annotated dataset that can be reused or extended for future plant stress detection studies [11].

With the rapid spread of cloud computing and mobile applications, this solution has strong potential to be scaled into easy-to-use mobile apps, where farmers can simply take a picture of a tea leaf and receive instant diagnostic feedback [12]. Ultimately, this innovation leads to sustainable tea production, empowers smallholder farmers with affordable tools, and strengthens food security in the face of climate change and resource depletion challenges in agriculture.

2. Background and literature survey

Tea is one of the important economies and socially beneficial large-scale cultivation crops in most countries. Tea plant health is extremely crucial to obtain high yields and quality. Tea plants are subject to a range of biotic stresses, such as fungal diseases (e.g., red rust, anthracnose) and insect pests (e.g., green leafhopper), and abiotic stresses, which comprise nutritional deficiencies (e.g., potassium deficiency) and abiotic damage (e.g., sunburn, drought stress).Both abiotic and biotic stresses are likely to produce visually similar symptoms on leaves, i.e., discoloration, yellowing, or lesions, and thus it is difficult for farmers to identify them manually. Misidentification results in inefficient management practices, e.g., the application of fertilizers instead of fungicides, resulting in loss of yield and economic loss. Thus, early stress identification in tea gardens is urgently needed using rapid, reliable, and cost-effective methods.

New imaging and artificial intelligence technologies have opened up new opportunities for non-destructive and high-throughput plant health monitoring. Multispectral and hyperspectral imaging have been employed to detect crop disease and nutrient deficiencies with relative success. These techniques are typically expensive, require controlled environments, and are not possible in small farmers' or students' experiments. On the other hand, RGB imaging using conventional cameras or smartphones enjoys the benefit of a low-cost, deployable-in-the-field, and scalable option, offering an attractive solution for practical agricultural applications.

2.1 Disease Detection in Tea Plants

Red Rust and Anthracnose: In previous studies, it has been shown that fungal diseases of tea leaves can be recognized using hyperspectral imaging in combination with machine learning or deep learning techniques. For instance, Huang et al. (2020) employed hyperspectral imaging and SVM for detecting fungal lesions on tea leaves with great precision in the laboratory setup.Insect Infestation: Green leafhopper detection in tea has primarily been done using visual inspection or standard RGB imaging, combined with CNN models such as YOLO or Faster R-CNN to detect insect infestation. Although these models are effective, they require extensive labeled datasets for their high performance.

2.2 Detection of Nutrient Deficiencies

Nutrient deficiencies, especially potassium deficiency, have been studied in plants like rice, maize, and grapevine using hyperspectral or multispectral imaging in combination with chemometric methods. Zhang et al. (2019) demonstrated that spectral indices can detect potassium deficiency in maize, but no such studies have been conducted on tea plants.RGB images have been less explored for nutrient deficiency detection due to subtle visible symptoms, even though recent studies show that deep learning can enhance differentiation even with normal images.

2.3 Deep Learning and Transfer Learning Application

Convolutional Neural Networks (CNNs) such as ResNet, EfficientNet, and MobileNet have been highly employed for the detection of plant stress under laboratory and field conditions.Transfer Learning facilitates the fine-tuning of pre-trained CNN architectures on limited datasets, which makes it amenable to student projects or small-scale field studies.Object detection frameworks like Faster R-CNN or Mask R-CNN have been utilized for localizing lesions on leaves, whereas segmentation methods (e.g., U-Net) measure the severity of the stress.

2.4 Limitations in Previous Work

Most studies focus on either disease detection or nutrient deficiency, not both, which is a key challenge for tea farmers who often misdiagnose visually similar stresses. Hyperspectral imaging studies are expensive and lab-based, limiting field application. RGB-based approaches are underexplored for tea plants, especially for distinguishing multiple stresses in real-field conditions.Few studies integrate lesion detection, severity quantification, and practical deployment, leaving a gap for low-cost, scalable solutions.

2.5 Opportunities

The use of low-cost RGB imaging combined with deep learning models provides a cost-effective, scalable, and viable solution for tea plant stress detection and discrimination. Transfer learning allows for accurate classification with relatively small datasets, making the strategy feasible for student-level research.

The strategy can be extended in the future to IoT-based continuous monitoring or UAV-based large-scale monitoring, bridging the gap between laboratory studies and actual agricultural practice.

3. Research Gap

Tea plants are highly susceptible to a wide range of stresses, including biotic factors like fungal infections (e.g., red rust and anthracnose) and insect infestations (e.g., green leafhopper), as well as abiotic factors such as nutrient deficiencies, drought, and sunburn. Early and accurate detection of these stresses is crucial for maintaining yield, quality, and overall plant health. Traditional diagnosis methods, primarily relying on visual inspection by farmers, are often subjective and prone to errors, especially when multiple stresses produce similar visual symptoms. For example, red rust and potassium deficiency both cause yellowing, discoloration, and lesions on leaves, making differentiation by the human eye extremely challenging. Misdiagnosis can lead to ineffective interventions, such as applying fertilizers instead of fungicides, which may further impact crop productivity and farm economics.

Previous studies have leveraged hyperspectral imaging (HSI), multispectral imaging, and laboratory-based analytical methods to detect diseases or nutrient deficiencies in tea and other crops. These approaches have demonstrated that spectral signatures and advanced machine learning models can accurately differentiate plant stresses. For instance, hyperspectral imaging combined with support vector machines or deep learning models has been successfully applied to classify fungal infections and nutrient deficiencies in controlled lab conditions. However, these methods present significant limitations: the equipment is expensive, datasets are typically collected under controlled lighting, and the systems are not readily deployable in the field. This makes them impractical for small-scale farmers or student research projects, limiting their real-world applicability.

While deep learning methods such as convolutional neural networks (CNNs) have been applied in plant stress detection, most studies focus on either disease detection or nutrient deficiency detection individually. Few studies consider the direct differentiation of biotic and abiotic stresses in tea plants, even though such differentiation is critical for accurate and timely management. Furthermore, existing deep learning studies often rely on large datasets or hyperspectral images, which are difficult to obtain and process for field conditions. RGB-based imaging, which is cost-effective, accessible, and portable, has been underexplored for this purpose, despite its potential for real-world applications.

Another significant limitation in previous research is the lack of integrated approaches that combine detection, localization, and severity quantification of stresses. Many studies stop at classification, providing only a label for an image or leaf without indicating the extent of damage or the affected area, which is crucial for informed decision-making by farmers. Similarly, very few approaches have been tested under realistic field conditions, where

variations in lighting, background clutter, and canopy structure can negatively impact model performance.

Taken together, these limitations indicate a clear research gap: there is a need for a practical, low-cost, RGB-based deep learning solution capable of accurately detecting and differentiating multiple visually similar stresses in tea plants. Such a system should ideally operate under field conditions, allow for lesion localization and severity quantification, and be scalable for future applications, including IoT-based continuous monitoring or UAV-assisted large-scale surveillance. Filling this gap will bridge the disconnect between advanced laboratory research and practical field implementation, providing a highly relevant and feasible solution for tea stress management that benefits both researchers and farmers.

In addition to the above gaps, it should be noted that tea, as a globally traded commodity and an important source of income for millions of smallholder farmers, calls for solutions not only precise but also low-cost, scalable, and easy to implement. Earlier research work, while scientifically worthful, has remained largely bogged down at the proof-of-concept stage and rarely translated into something that is easily implementable by farmers on the tea estates. Excessive reliance on controlled laboratory datasets, costly sensors, or advanced preprocessing pipelines creates an interface imbalance between what can be accomplished scientifically and what can practically be implemented in tea gardens. Additionally, most existing models fail to capture the progressive and temporal nature of stresses—how initial signs of potassium deficiency or fungal infection evolve through different visual stages that confuse farmers as much as they confuse algorithms trained on limited data. This is addressed by models that are adaptive, robust, and generalizable to broad field conditions. Another essential component that is lacking in existing studies is the integration of decision-support features. While classification of stress types is important, farmers also need actionable information—estimated severity, recommended interventions, and likely economic consequences—to inform management decisions. Current studies hardly bridge this gap between scientific diagnosis and farmer decision-making. Similarly, little research has been done on establishing open-access RGB-based datasets for tea stresses that are important for reproducibility, benchmarking, and encouraging further innovation in the field.

4. Research Problem

Tea is one of the most economically important crops worldwide, and its productivity and quality are highly dependent on the health of the plants. Tea plantations are frequently affected by a combination of biotic stresses, such as fungal infections like red rust and anthracnose, and abiotic stresses, including nutrient deficiencies such as potassium deficiency, drought stress, or leaf sunburn. These stresses often produce similar visual symptoms, including yellowing, reddish or brown lesions, chlorosis, and necrosis, which are difficult for farmers and plantation workers to differentiate through visual inspection alone. Inaccurate or delayed diagnosis can lead to improper interventions, such as applying fertilizers when a fungicide is required, resulting in yield loss, economic inefficiency, and exacerbation of plant stress. Current research has shown that advanced imaging techniques, such as hyperspectral and multispectral imaging, combined with machine learning models, can effectively detect plant stresses. These techniques exploit subtle spectral differences in leaves to differentiate diseases from nutrient deficiencies or other abiotic stresses. While highly accurate, these methods are costly, equipment-intensive, and often limited to controlled laboratory environments, making them largely inaccessible to smallholder farmers, student researchers, or large-scale field deployment in remote tea plantations.

Additionally, most existing studies focus on either disease detection or nutrient deficiency detection individually. Few studies attempt to address the direct differentiation between multiple visually similar stresses in tea plants, despite the real-world relevance of this challenge. There is also a scarcity of research on RGB image-based approaches, which are far more affordable and practical for field applications compared to hyperspectral systems. Moreover, prior works often emphasize classification alone, without localizing lesions, quantifying severity, or providing actionable insights for farmers.

This highlights a significant gap in practical agricultural research: there is a need for a robust, cost-effective, and scalable system that can leverage affordable RGB imaging and deep learning to detect, differentiate, and quantify multiple stresses in tea plants under real-world field conditions. Such a system would not only enable early detection but also provide quantitative information on stress severity, supporting informed decision-making and precise intervention. Therefore, the central research problem for this study can be formulated as follows: How can a low-cost, RGB image-based deep learning system be designed and implemented to accurately detect, differentiate, and quantify visually similar biotic and abiotic stresses in tea plants under varying real-world field conditions, while remaining practical and scalable for use by farmers and researchers, Addressing this problem will bridge the gap between expensive laboratory-based stress detection methods and affordable, practical solutions that can be applied directly in tea plantations. The study aims to provide a student-friendly, deployable approach using widely available tools and equipment, demonstrating that high-accuracy stress detection can be achieved without relying on costly

hyperspectral imaging, while laying the foundation for future integration with IoT devices, UAV monitoring, and large-scale agricultural applications.

Earlier research in detecting stress in tea plantation has been a good source of learning about abiotic and biotic stresses. A more detailed evaluation, however, raises a series of major shortcomings that restrict their applicability in real-world field conditions. Most of the studies presented have a limited scope, addressing detection of a single agent of stress at a time. For instance, whereas some research is concentrated on the identification of fungal diseases such as red rust (caused by Cephaleuros spp.), others are specifically aimed at abiotic stress conditions like potassium deficiency or nitrogen imbalance. Although such techniques provide useful information under controlled situations, they prove ineffective in addressing the multifactorial nature of tea gardens. In practice, plants normally suffer from a number of stresses simultaneously or consecutively such that growers must deal with cumulative signs that are much more subtle than in idealised models. A second major problem is that signs of different stresses are extremely difficult to distinguish visually. For example, early symptoms of potassium deficiency, such as yellowing and spotting on leaves, may closely resemble the early phases of fungal or algal infection. This overlap contributes to diagnostic uncertainty and postponements in the implementation of correct remedies. Farmers, who are generally bereft of facilities for advanced laboratory work, must rely on hand inspection and personal judgment, both of which have low accuracy rates. Misdiagnosis here not only jeopardizes recovery but also tends to increase the issue by applying the incorrect fertilizer or pesticide. Thus, whereas recent studies offer useful technical information, failure of these studies to adequately distinguish more than a few and confounding stressors leads to a severe lacuna in the issue of practical application. In addition, most of state-of-the-art detection equipment relies on hyperspectral imaging (HSI) or multispectral imaging (MSI) techniques. While useful in the laboratory, these methods are hindered in big tea gardens by being costly machinery, heavy equipment, and sophisticated data processing requirements. For small tea producers, especially in the developing world, neither being sustainable nor affordable is it to use such technologies. It created a gap between lab-based innovations and the practical users' needs who require cheap, reliable, and easy-to-use devices. These gaps underscore the need for an improved more integrated, efficient, and low-cost means of identifying stresses in tea plants. Although HSI or MSI methods might not be practical, RGB imaging is an available option. High-resolution RGB images can be taken using inexpensive cameras and recent smartphones and identified based on machine learning algorithms to differentiate between stresses. This process not only decreases costs significantly but also democratizes access, allowing smallholder farmers to access advanced diagnostic technologies with minimal capital investment. Additionally, RGB-based systems can be augmented using artificial intelligence to increase precision in detecting subtle changes that may be difficult for the human eye to see.

5. Objective

5.1 Main Objectives

To design and implement a low-cost, RGB image-based deep learning system capable of accurately detecting, differentiating, and quantifying multiple visually similar biotic and abiotic stresses in tea plants under real-world field conditions. The system aims to provide actionable insights for tea farmers and plantation managers, enabling timely and effective stress management, improving yield and quality, and serving as a foundation for scalable, automated monitoring solutions.

5.2 Specific Objectives

1. **Data Collection and Dataset Development**
    - To capture a diverse and representative dataset of tea leaf images in real-field conditions, covering healthy leaves as well as leaves affected by biotic stresses (e.g., red rust, anthracnose) and abiotic stresses (e.g., potassium deficiency, sunburn).
    - To ensure diversity in the dataset by including variations in lighting conditions, canopy levels, leaf orientation, and environmental backgrounds.
    - To annotate and label images accurately for classification tasks (healthy vs stressed) and, optionally, for lesion detection and segmentation to quantify severity.

2. **Image Preprocessing and Enhancement**
    - To implement preprocessing steps such as resizing, normalization, noise reduction, and contrast enhancement to improve image quality.
    - To apply data augmentation techniques including rotation, flipping, brightness and contrast adjustments, and slight colour variations, enhancing the model's ability to generalize across diverse field conditions.

3. **Deep Learning Model Development**

    - To leverage transfer learning on pre-trained CNN architectures such as ResNet50, MobileNetV2, or Efficient Net for efficient training on a limited dataset while achieving high accuracy.
    - To explore object detection and segmentation models like Faster R-CNN, Mask R-CNN, or U-Net for precise lesion localization and assessment of stress severity.
    - To integrate classification and lesion quantification in a stepwise workflow, allowing both stress identification and measurement of affected areas.

4. **Model Evaluation and Validation**

    - To evaluate model performance using standard metrics, including accuracy, precision, recall, F1-score, and confusion matrix, and compare the results with manual visual assessments.
    - To test the robustness of the model under different lighting conditions, leaf orientations, and backgrounds to ensure real-field applicability.
    - To analyse cases of misclassification or detection errors to identify areas for future improvement.

5. **Practical Application and Field Deployment**

    - To demonstrate the feasibility of the developed system using low-cost RGB cameras or smartphones, making it accessible to students, researchers, and smallholder farmers.
    - To propose a workflow for future deployment with IoT-based sensors or UAV-mounted cameras, enabling automated, large-scale monitoring of tea plantations.
    - To provide insights and recommendations for early stress detection and intervention, helping farmers make informed decisions on fertilizer or fungicide application.

6. **Contribution to Research and Industry**

- To fill the existing gap in RGB-based deep learning approaches for differentiating multiple visually similar stresses in tea plants, which is currently underexplored.
- To develop a student-friendly, low-cost methodology that can be used for research, academic projects, or practical agricultural applications.
- To lay the groundwork for further studies that integrate real-time monitoring, IoT, or UAV-based systems, advancing precision agriculture in tea cultivation.

6 Methodology

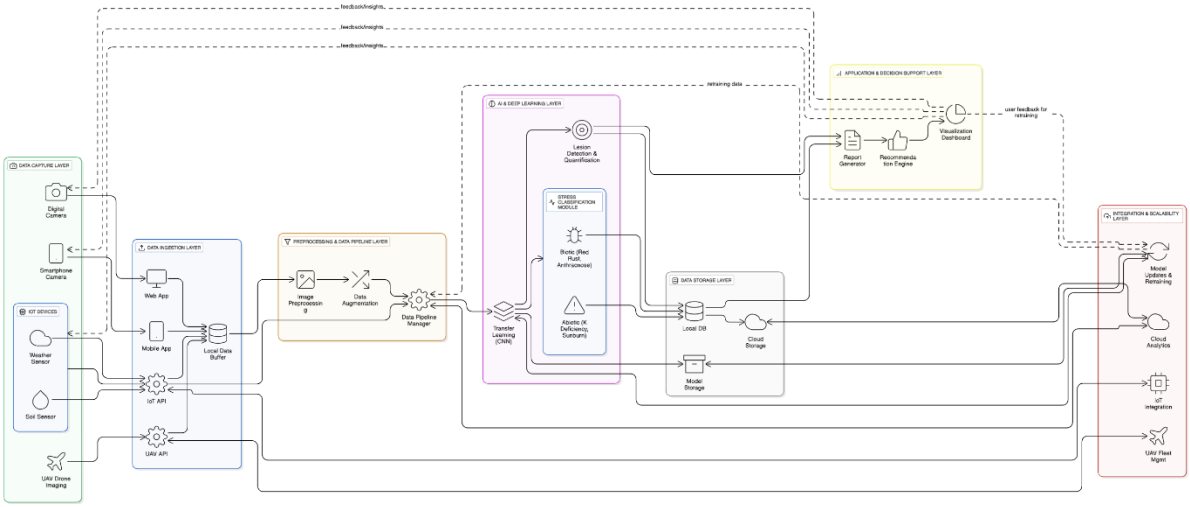


Figure 1: Architecture diagram

The proposed methodology aims to develop a low-cost, RGB image-based deep learning system for the detection, differentiation, and quantification of biotic and abiotic stresses in tea plants. This approach leverages widely available resources such as smartphone or digital cameras, making it highly accessible for students, researchers, and smallholder farmers. The methodology is designed to function under real-field conditions, accommodating natural variations in lighting, leaf orientation, canopy structure, and background complexity, which are often limitations in controlled laboratory-based studies. The methodology integrates computer vision techniques, image preprocessing, data augmentation, and deep learning models to achieve accurate classification of tea leaves into multiple stress categories, including biotic stresses like red rust and anthracnose, and abiotic stresses such as potassium deficiency and leaf sunburn. By combining transfer learning with pre-trained convolutional neural networks (CNNs), the system can achieve high accuracy even with a relatively small dataset, making it feasible for student-led research. Furthermore, the methodology incorporates optional lesion detection and segmentation to quantify the severity of stresses, providing actionable insights for farmers. The workflow emphasizes a practical and scalable approach, enabling future integration with IoT devices, UAV-based monitoring, or large-scale automated tea plantation surveillance. By focusing on affordable RGB imaging and open-source tools, this methodology bridges the gap between high-cost laboratory techniques and

field-deployable solutions, offering a robust, non-invasive, and high-throughput system for early stress detection and management in tea cultivation.

The proposed methodology revolves around the development and design of a low-cost, RGB image-based deep learning system capable of identifying, discriminating, and counting both abiotic and biotic stresses in tea plants with the eventual target of offering a realistic, economical, and field-deployable platform for students, scientists, and small farmers. In contrast to traditional lab-based methods involving costly hyperspectral or multispectral imaging technologies, this method relies on widely accessible hardware like smartphones and digital cameras, making it possible to adopt economically by various end users. The fundamentum of this method lies in its capability to work effectively in genuine field conditions, under which environmental components such as varying light intensities, shadow, leaf orientation, canopy density, ground reflection, and highly structured backgrounds typically hinder precise image acquisition and analysis. By combining cutting-edge computer vision techniques, intelligent preprocessing pipelines, and persistent deep learning models, the proposed method promises resilience against such natural variations and thereby is viable for direct deployment on tea estates irrespective of artificially controlled conditions. The foundation of the method is the application of convolutional neural networks (CNNs), i.e., transfer learning methods with pre-trained architectures such as ResNet, VGG, or EfficientNet, which will enable the system to learn properly even when having a relatively small dataset size, as is normally the case in resource-limited or student projects. These pre-trained networks make a good feature extraction foundation, which, after fine-tuning with domain-specific images of tea leaves, can be utilized to differentiate abiotic stresses such as potassium deficiency and leaf sunburn from biotic stresses such as red rust and anthracnose. For making the models stronger and more generalizable, the method gives strong emphasis on systematic data preprocessing and data augmentation techniques such as rotation, flipping, brightness normalization, background subtraction, and synthetic noise addition, which collectively simulate real-field variability and minimize the risks of overfitting. In addition to classification, the method also throws its gaze towards voluntary lesion detection and segmentation using pixel-level deep learning architectures like U-Net or Mask R-CNN to detect and quantify the intensity of stress lesions. This not only enables categorical classification of stresses but also quantifies severity, thus generating actionable information for farm management decision-making such as pesticide spraying schedules, nutrient enrichment, or canopy management regimes. Moreover, this quantification of functionality in lesions enhances the system interpretability, providing the farmers with a straightforward visual representation of disease spread or extent of nutrient shortage, which builds trust in the machine suggestions.

Besides the technical platform, the solution aims for scalability, flexibility, and compatibility with future agricultural technologies. For instance, once validated with handheld image

capture devices, the same platform can readily be integrated with UAV (unmanned aerial vehicle) or drone-based systems for monitoring large areas over extensive tea estates, or IoT-enabled devices for real-time monitoring and automated alerts. Such scalability renders the methodology not just suitable for smallholder farms but also potentially able to support industrial-scale tea farming, where stress early detection significantly reduces crop loss and input wastage. The affordability focus is continued by using only RGB imaging and open-source software tools such as TensorFlow, PyTorch, and OpenCV so that the method avoids costly proprietary solutions and is as accessible as possible to students and research teams in the developing world. The pipeline is designed to be easy to use, where smartphone photos can be uploaded to a light app or cloud server, processed in near real time, and returned with stress detection and severity outcomes. Furthermore, the method promotes a cycle of continuous improvement, where feedback from in-field users in the form of misclassifications or edge cases is incorporated in iterative model retraining, thereby making the system more reliable and adaptable with time. Ethical concerns, such as data privacy upon image upload and freedom from biases by portraying diverse types of stresses across locations, are also encompassed within the method. In effect, this approach bridges the gap between lab-based, high-expense solutions and field-deployable, practical systems by introducing a high-throughput, non-invasive, and robust detection pipeline to enable farmers to take early action in tea crop management. By facilitating the early identification of stress and severity estimation at scale, this method not only avoids yield loss but also helps in sustainable agricultural practices by reducing the application of unnecessary pesticides, conserving resources, and promoting precision agriculture in tea cultivation. Finally, the approach stands as a revolutionary innovation in agricultural technology, one that holds scientific integrity and practitioner usability in equal regard, creating an ecosystem in which technology, farmers, and scientists work together toward healthier crops, better productivity, and more sustainable livelihood.

The proposed methodology seeks to establish and implement a low-cost, RGB image-based deep learning system for identification, differentiation, and quantification of both abiotic and biotic stresses affecting tea plants. This technology is being propelled by the exigent demand for affordable and efficient plant stress identification technologies that could be applied not just by scientists and commercial tea estates but also by students and smallholder farmers, who generally lack the capital to invest in expensive laboratory-based or multispectral imaging technologies. Making use of the ubiquity of digital cameras and smart phones, the method circumvents the utilization of sophisticated and costly sensing hardware and thus offers the benefits of scalability and convenience. The system is intended to work optimally in real-world field settings where environmental heterogeneity is inevitable—natural lighting that changes, tree canopy shadows, disordered leaf orientations, background clutter due to

soil or weeds, and weather-like scenarios such as wet leaves. Unlike traditional laboratory settings, where images are captured over uniform backgrounds and altered under controlled lighting, this method embraces variability and obtains resilience with properly designed preprocessing, augmentation, and model adaptation strategies to ensure the proposed method does not collapse when deployed outside the academic world.

The method employs a multi-stage pipeline which begins with RGB camera-based data collection. Photos of tea leaves are shot under different conditions—times of day, weather, camera position and distance—to produce a representative dataset approximating the conditions in the field. These images contain diverse forms of stresses, such as biotic stresses such as red rust (caused by Cephaleuros app.), anthracnose, and leaf blight, and abiotic stresses such as potassium deficiency, nitrogen deficiency, and sunburn. Acquiring such a data set requires a planned effort, entailing collaboration with tea estates, agricultural research centers, and even crowdsourcing on farmers' networks and students' contributants who can share images on a shared repository. Raw data is then passed under a strict image preprocessing pipeline in which images are resized to typical dimensions suitable for deep learning input, background noise is reduced through filtering mechanisms, and color balancing operations are performed to remove variations caused by illumination. Data cleaning also becomes a necessary operation where faulty, blurred, or duplicate images are eliminated to ensure dataset purity. After preprocessing, the data augmentation is employed in order to synthetically increase dataset diversity and size, which deep learning models typically require big datasets to learn. Techniques such as random rotation, flipping, scaling, contrast adjustment, brightness modification, and addition of Gaussian noise are employed for simulating real-world variability and increasing the ability of the system to generalize. This enhancement assists in avoiding the overfitting of the trained model into regulated environments and can rightfully identify stresses under unfamiliar environmental conditions.

6.1 Software Solution

- **Python programming language**, which is open-source and widely used in machine learning and computer vision.
- **Deep learning frameworks**: TensorFlow or PyTorch for building, training, and evaluating CNN models.
- **OpenCV** for image processing tasks including preprocessing, filtering, and augmentation.
- **Label Image or similar tools** for image annotation and creating labeled datasets.
- **Spreadsheet or database software** for organizing image metadata, labels, and experimental results.

3.1.6 Use Case Diagram

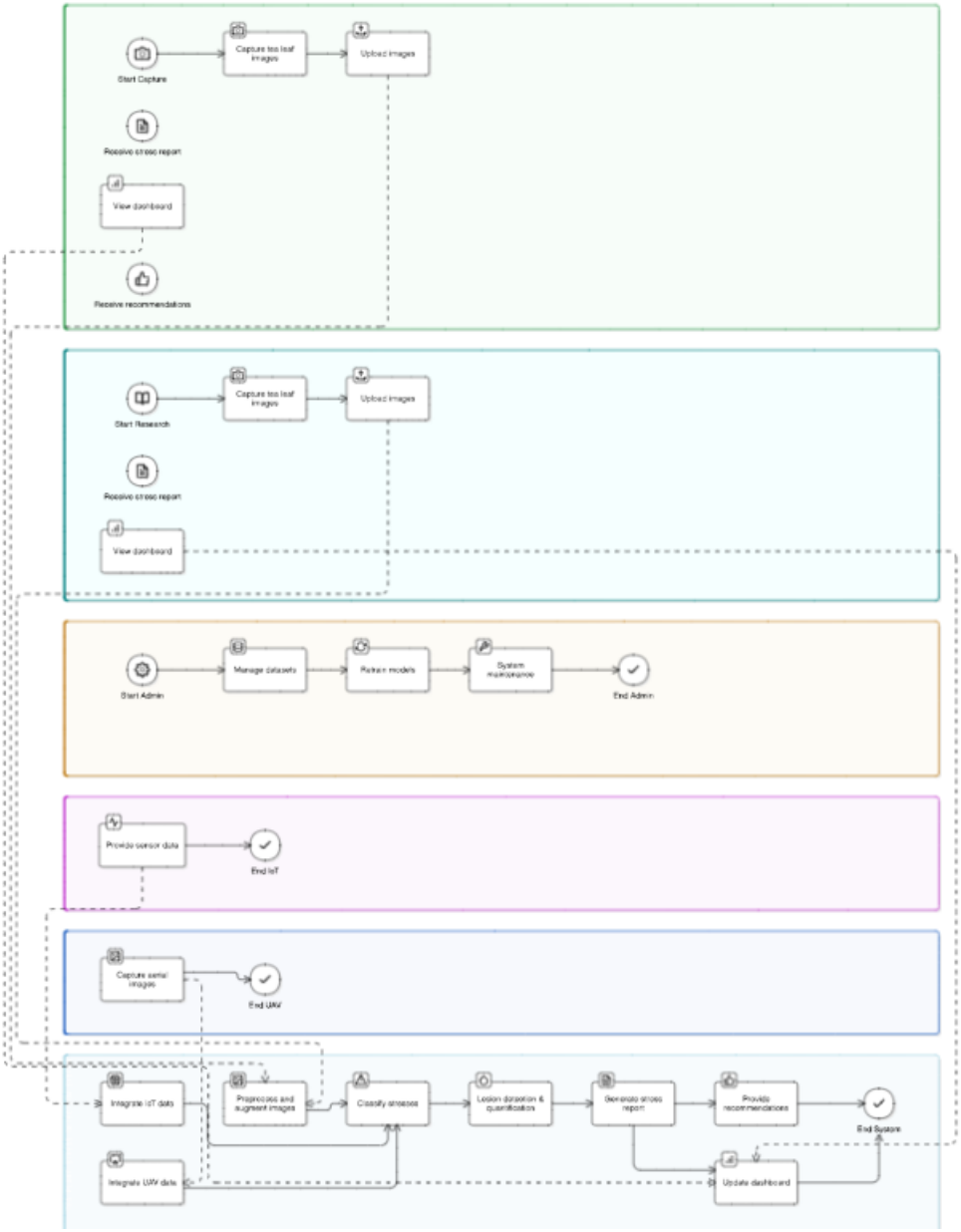


Figure 2:use case diagram

3.1.7 Sequence Diagram

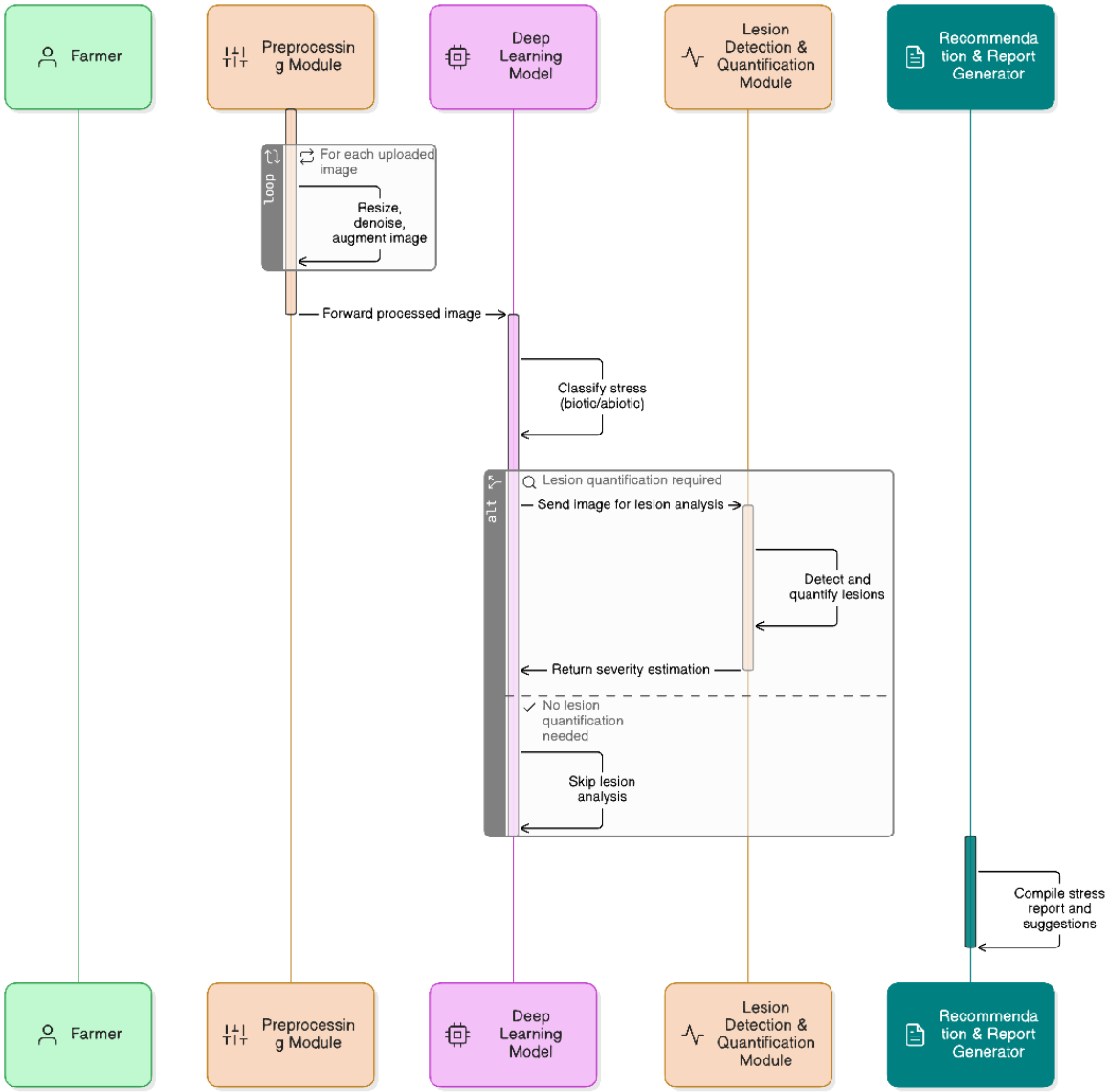


Figure 3:Sequence diagram

7 User Requirements

1.Functional User Requirements

1. Users can capture RGB images of tea leaves using a smartphone or digital camera.
2. Users can upload pre-captured images for analysis.
3. Users can submit images for stress classification with a single click.
4. Users can view results, including:
    - Stress type (e.g., healthy, red rust, potassium deficiency)
    - Confidence score of classification
    - Optional highlighted lesion areas with severity percentage
5. Users can save or export results for documentation or further analysis.

2. Usability Requirements

1. System should have a simple, intuitive interface suitable for students or non-technical users.
2. Instructions for image capture, upload, and analysis should be clear and easy to follow.
3. System should provide feedback or alerts in case of invalid inputs (e.g., non-leaf images).

3. Performance Requirements for Users

1. Classification results should be generated quickly (ideally within 5 seconds per image).
2. The system should allow batch uploads or sequential analysis without crashing.

4. Accessibility Requirements

1. Users should be able to access the system on laptops or cloud platforms.
2. The system should require minimal technical setup, relying mostly on web-based or pre-configured environments.

4.2 Software Requirements

1. Programming Language
    - Python 3.x: Core programming language for data preprocessing, model training, and image analysis.
2. Deep Learning Frameworks
    - TensorFlow or PyTorch: For implementing convolutional neural networks (CNNs) and transfer learning models for stress classification.
    - Optional: Keras (high-level API in TensorFlow) for simplified model building and experimentation.
3. Image Processing Tools
    - OpenCV: For preprocessing images (resizing, normalization, color adjustment, noise reduction).
    - NumPy / Pandas: For data manipulation and handling metadata.
    - Matplotlib / Seaborn: For visualization of results, graphs, and model performance metrics.
4. Annotation Tools
    - LabelImg or RectLabel: For labeling images and optionally drawing bounding boxes for lesion detection.
5. Development Platforms
    - Google Colab / Kaggle Notebooks: Cloud-based platforms providing free GPU support for model training, making it accessible for students without high-end hardware.
    - Local Python IDEs such as PyCharm, VS Code, or Jupyter Notebook for development and debugging.
6. Optional Tools
    - TensorBoard: For monitoring training progress and visualizing model metrics.
    - Cloud Storage (Google Drive, Dropbox): For storing datasets and model checkpoints.

4.3 Functional Requirements

1. Image Capture

- Capture RGB images of tea leaves using a smartphone or digital camera.
- Allow uploading of existing leaf images to the system.

2.Image Preprocessing

- Resize, normalize, and enhance images.
- Apply data augmentation (rotation, flipping, brightness/contrast adjustment).

3.Stress Detection and Classification

- Detect tea leaves in images.
- Classify leaves into healthy, red rust, potassium deficiency, or other stresses.

4.Lesion Detection and Severity Estimation (Optional)

- Detect affected leaf areas using segmentation or bounding boxes.
- Quantify severity by calculating affected area percentage.

5.Results Display

- Show classified stress type with confidence scores.
- Display images with highlighted lesions if lesion detection is enabled.

6.Data Storage and Management

- Store images, annotations, and prediction results for future analysis.

4.4 Non-Functional Requirements

1.**Performance**

Predictions must be generated in less than 5 seconds per image (depending on hardware).

2.**Scalability**

System should handle hundreds of images without performance degradation.

3.**Usability**

User interface should be simple and intuitive, suitable for students and farmers.

4.**Portability**

Should run on laptops or cloud platforms (e.g., Google Colab).

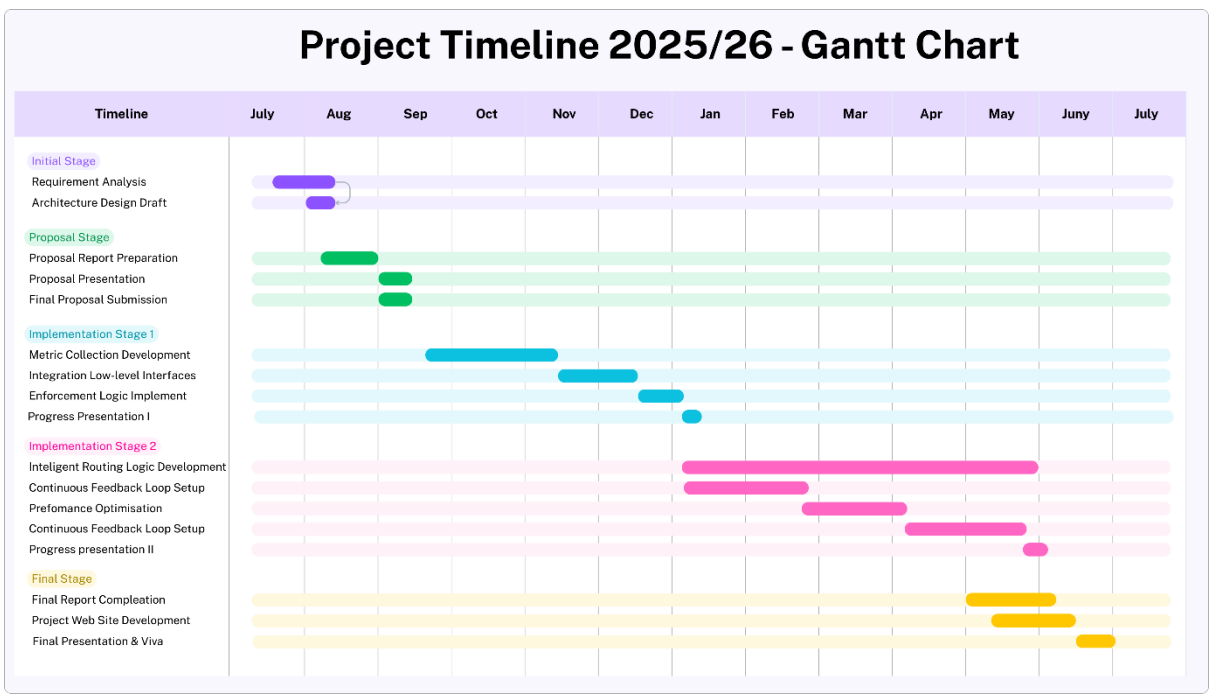
5.**Reliability**

System should maintain accuracy above 85**%** under varying field conditions.

6.**Maintainability**

Easy to retrain or update the model with new images or stress types.

Gantt Chart



Work breakdown Chart

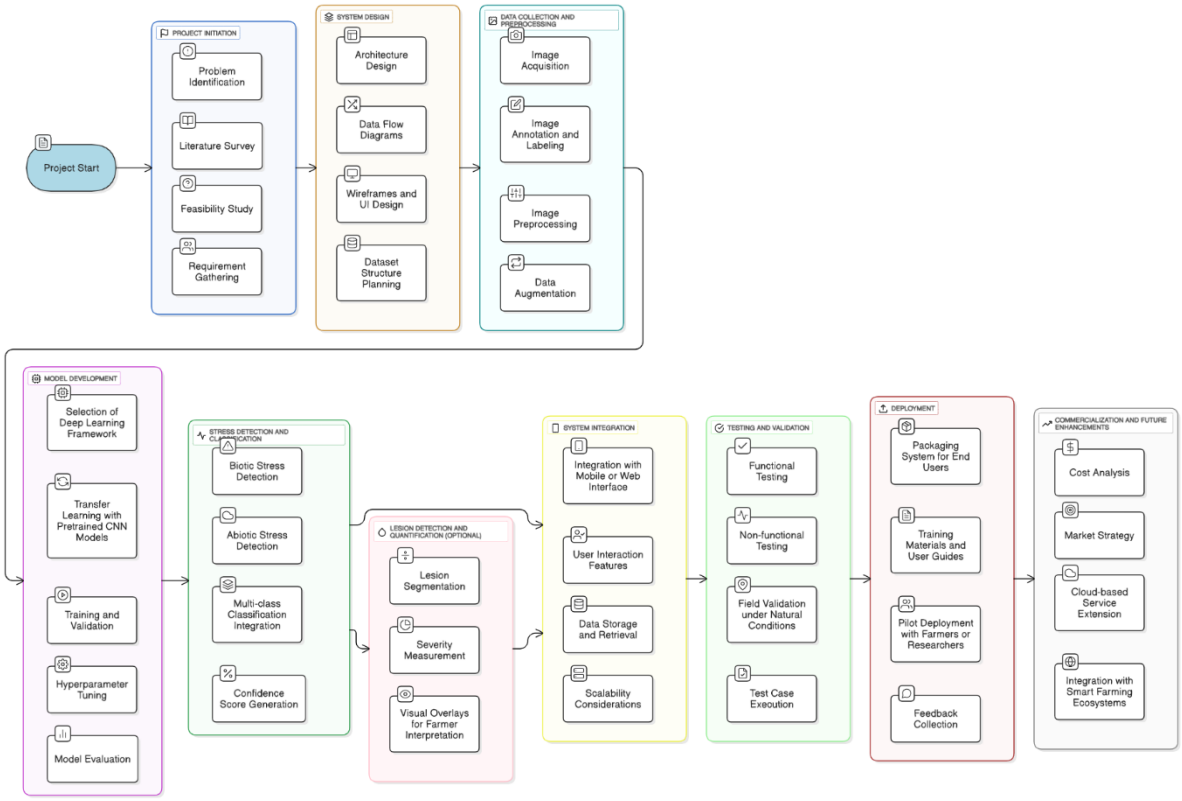


Figure 4:work breackdown chart

Justification of Facilities

The combination of personal skills and available facilities ensures that the research can be conducted efficiently without relying on costly laboratory equipment. The student-level approach using affordable RGB imaging, open-source software, and cloud-based computation allows for:

- Practical and real-field data collection.
- Efficient model training and evaluation.
- Scalable methodology that can be extended to larger plantations or future IoT/UAV integration.
- A non-invasive and eco-friendly approach to monitoring tea plant health.

8 Budget and budget justification

The proposed study is designed to be **low-cost and student-friendly**, leveraging affordable resources such as RGB cameras (smartphones), open-source software, and cloud computing platforms. The budget focuses on essential items required for **data collection, model development, and analysis**, ensuring practical feasibility for a student-led research project.

Table 1: Budget

Component	Cost
Travelling Cost	Rs. 5,000
Deployment Cost	Rs. 5000/month
Web app hosting	Rs. 4000
Cloud computing cost	Rs. 8000

Budget Justification

1. **Smartphone / Digital Camera**
    - Essential for capturing RGB images of tea leaves in natural field conditions.
    - Affordable and portable, enabling collection under varying lighting, angles, and backgrounds.
2. **Laptop / Personal Computer**
    - Necessary for data preprocessing, model training, and evaluation.
    - GPU support is beneficial for deep learning but not mandatory, as cloud GPU platforms can be used.
3. **Data Storage**
    - Large image datasets require reliable storage for both training and backup.
    - An external hard drive or cloud storage ensures data security and easy access.

4. **Cloud Computing / Internet Costs**
    - Training deep learning models locally may be time-consuming without a GPU.
    - Cloud services such as Google Colab Pro provide free or low-cost GPU access for faster training.
5. **Software**
    - All software tools required for this project (Python, TensorFlow, OpenCV, LabelImg) are open-source, minimizing cost.
    - These tools are widely used in both research and industry, providing a practical learning experience for students.
6. **Miscellaneous Costs**
    - Minor expenses for field visits, transportation, and printing.
    - Necessary to ensure smooth data collection and documentation.

9 Commerlization

The proposed RGB image-based deep learning system for tea plant stress detection has strong potential for commercialization, particularly in the agriculture technology (AgriTech) sector. Its key strengths—low-cost, accessibility, and ease of use—make it attractive to smallholder farmers, tea estates, agricultural extension services, and research institutions.

1. Target Market

1. **Tea Plantations and Estates**
   - Large-scale tea growers can use the system for early detection of biotic and abiotic stresses, improving yield and quality.
2. **Smallholder Farmers**
   - Affordable and smartphone-compatible, allowing non-expert farmers to monitor crop health without hiring agronomists.
3. **Agricultural Advisory Services**
   - Extension officers can integrate the system into field scouting operations to provide real-time guidance.
4. **Research and Educational Institutions**
   - Universities and research centers can use the system for student projects, plant pathology studies, and precision agriculture research.

2. Product Variants

1. **Mobile App Version**
   - User-friendly smartphone application that allows farmers to capture leaf images, get instant stress detection results, and receive recommended interventions.

2. **Web-Based Dashboard**
    - For tea estates or research centers to upload batches of images, track stress patterns, and monitor multiple plantations remotely.
3. **IoT/Drone Integration (Future Expansion)**
    - Advanced variant integrating drones or IoT devices to scan entire plantations and provide real-time monitoring of tea crop health.

3. Revenue Models

1. **Subscription-Based Model**
    - Monthly or annual subscription for premium features such as batch analysis, historical data tracking, or drone integration.
2. **One-Time Purchase / License Fee**
    - Users can buy the software for offline use with basic features like single-image analysis and stress classification.
3. **Freemium Model**
    - Free access to basic stress detection, with paid add-ons for severity quantification, lesion detection, and cloud storage.

4. Marketing Strategy

1. **Partnerships with Tea Associations**
    - Collaborate with tea boards or farmer cooperatives to distribute the app and conduct workshops.

2. **Online Marketing & AgriTech Expos**
    - Showcase the system at agriculture technology fairs, webinars, and social media platforms targeting farmers and agronomists.
3. **Demonstration Projects**
    - Pilot studies with local tea estates to demonstrate the cost-effectiveness and accuracy of the system.

5. Competitive Advantage

1. **Low-Cost & Accessible**
    - Unlike hyperspectral imaging or lab-based solutions, the system works with standard RGB cameras, making it affordable for students and farmers.
2. **Non-Invasive & Easy to Use**
    - Requires minimal technical skills for operation, with automated classification and results display.
3. **Scalable & Future-Ready**
    - Can be extended to drone/UAV monitoring, IoT integration, or multi-crop stress detection, making it a versatile AgriTech solution.

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